A Low-Cost Single-Probe Differential-pH Aeration Monitor for Seawater Carbonate Alkalinity: Systematic-Error Decomposition and Sub-0.25 dKH Accuracy via Sealed Common-Headspace Equilibration

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June 22, 2026

Abstract

Carbonate alkalinity ("KH", expressed in degrees, dKH) is the single most control-critical water-chemistry parameter in reef aquaria, yet affordable continuous monitors are scarce and reagent titration is laborious. We describe an open, reagent-free monitor that infers alkalinity from the equilibrium pH a seawater sample reaches after aeration to atmospheric CO_2, using a *single* pH probe to measure an aquarium sample ("tank") and a known-alkalinity reference ("ref") in immediate succession. Because dKH is computed from the pH *difference* ($\mathrm{KH_{tank}} = \mathrm{KH_{ref}} \cdot 10^{\Delta \mathrm{pH}}$), absolute probe drift and room-CO_2 excursions are common-mode signals that cancel. Over a nine-day instrumented campaign (29 logged measurements across five firmware/method revisions) we observed a persistent uncorrected error of -0.92 dKH and, rather than zero-point calibrating it away, decomposed it. Three interventions—(i) plateau-following measurement to true gas equilibrium, (ii) a single *sealed common headspace* shared by sample, reference and pump intake, and (iii) an actively aerated 5 L reference anchor that pins the headspace $p_{\mathrm{CO_2}}$—collapsed the dawn-CO_2 error and the diurnal evaporation confound, and shrank measurement scatter to SD $= \pm 0.08$ dKH (about one-third of the ± 0.25 dKH design goal). A sequence of null experiments (identical water in both vessels, true $\Delta\mathrm{pH} = 0$) then localized the residual: a *complete null* returned $\Delta\mathrm{pH} = -0.003$, dKH $= 8.387$, i.e. an instrument offset of only -0.06 dKH. The remaining -0.86 dKH was traced to alkalinity loss (CO_2 driven $CaCO_3$ precipitation / biofilm) in the days-old, small, open tank aliquot—a sample-handling artifact, not a measurement fault. A re-run of the null on the same aged aliquot a day later showed the offset decaying from -0.06 to -0.22 dKH with the electronics untouched, confirming the attribution in the time domain. The instrument therefore meets the ± 0.25 dKH goal uncorrected, with zero-point calibration deemed unnecessary. Our independently obtained error structure (room-CO_2 sensitivity, bubble-quality dependence, sealing as the remedy) reproduces the published behaviour of a commercial counterpart, providing external validation of both the diagnosis and the remedy.

Keywords: reef aquarium; carbonate alkalinity; total alkalinity; CO_2 equilibration; differential pH; low-cost instrumentation; common-mode rejection; $CaCO_3$ precipitation.

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1 Introduction

Stony corals and other calcifying reef organisms deposit calcium carbonate at a rate that continuously consumes carbonate alkalinity. In a closed reef aquarium the resulting alkalinity drawdown is fast (often 0.5–1.5 $\mathrm{dKH\,day^{-1}}$) and swings of more than $\sim 1\,\mathrm{dKH}$ are implicated in coral tissue stress. Alkalinity (A_T, conventionally reported by hobbyists as "KH" in German degrees, $1\,\mathrm{dKH} \approx 0.357\,\mathrm{meq\,L^{-1}}$) is therefore the parameter most worth measuring *often*.

The accessible options are unsatisfying for continuous use. Manual reagent titration (e.g. commercial drop kits) is accurate but labour-intensive and discrete. Colorimetric robots automate titration but consume reagent and are expensive. Direct potentiometric pH alone is *not* an alkalinity measurement, because seawater pH depends jointly on A_T and $p_{\mathrm{CO_2}}$, and aquarium $p_{\mathrm{CO_2}}$ varies strongly over the day (room ventilation, photosynthesis, respiration).

A reagent-free alternative exploits the fact that if a sample is aerated until its $p_{\mathrm{CO_2}}$ equilibrates with the surrounding air, the equilibrium pH becomes a single-valued function of alkalinity at fixed temperature and salinity. A commercial product (AquaWiz) is built on this principle and is used by reef hobbyists, but community reports document recurring inaccuracy tied to room-CO_2 fluctuation and to aeration/bubble quality. The contribution of this paper is not the principle but a *quantitative diagnosis*: starting from a working DIY build, we conduct a nine-day instrumented campaign, expose a $-0.92\,\mathrm{dKH}$ raw error, and show by construction (null experiments) that essentially all of it is a sample-handling artifact rather than an instrument limitation—reducing the true instrument offset to $-0.06\,\mathrm{dKH}$ and meeting a $\pm 0.25\,\mathrm{dKH}$ goal without calibration. Along the way we establish a small set of design rules (sealed common headspace; plateau-following to true equilibrium; an actively aerated reference anchor; a two-gate plateau criterion) that we expect to generalize to any single-probe differential-pH alkalinity monitor.

2 Measurement Principle

2.1 Aeration makes equilibrium pH a proxy for alkalinity

At fixed temperature T and salinity S, the seawater carbonate system has two degrees of freedom; fixing any two of $\{A_\mathrm{T}, \mathrm{DIC}, p_{\mathrm{CO_2}}, \mathrm{pH}\}$ determines the rest. Aeration with a large excess of room air drives the dissolved CO_2 toward Henry's-law equilibrium with the gas phase, $[\mathrm{CO_2}]_{aq} = K_H\, p_{\mathrm{CO_2}}$, so that $p_{\mathrm{CO_2}}(\text{water}) \to p_{\mathrm{CO_2}}(\text{air})$. Once $p_{\mathrm{CO_2}}$ is clamped by the atmosphere, the equilibrium pH is a single-valued function of alkalinity:

$$\mathrm{pH_{eq}} = f(A_\mathrm{T},\, p_{\mathrm{CO_2}},\, T,\, S). \tag{1}$$

With $p_{\mathrm{CO_2}}$, T, S held common, the *only* variable that can move $\mathrm{pH_{eq}}$ is A_T. Measured equilibrium pH is therefore a proxy for alkalinity.

2.2 Why the differential cancels room CO_2

The instrument never relies on the absolute value of $p_{\mathrm{CO_2}}$. It aerates the reference and the tank sample to the *same* atmospheric ceiling (in the final design, by aerating both against one shared headspace simultaneously, Sec. 4) and reports the *difference* of their equilibrium pH. Because the absolute $p_{\mathrm{CO_2}}$ term is common to both samples, it cancels in the difference: a room-CO_2 excursion is a common-mode signal that drops out. What survives the subtraction is the true alkalinity difference plus electrode noise (± 0.005–$0.01\,\mathrm{pH}$). This is the central robustness argument of the method, and the reason a *single* probe (rather than two matched probes) is used: a single sensor's slow asymmetry-potential and reference-junction drift load identically onto both readings and cancel in ΔpH.

2.3 Over-aeration is impossible; only *insufficient* aeration errs

Equation (1) has a hard ceiling at p_{CO_2}(air): once the water reaches atmospheric equilibrium, further aeration cannot raise pH. There is consequently no "over-aeration" failure mode. Error enters only through *insufficiency*: if residual CO_2 remains in one vessel, that side's pH is unfairly depressed and a spurious ΔpH appears. The differential's assumption is precisely that ref and tank share the same p_{CO_2}; broken symmetry breaks the cancellation. This motivates measuring to a genuine plateau rather than for a fixed time (Sec. 4).

2.4 Volume independence at full equilibrium

Equation (1) contains no volume term (Henry's law is intensive). Hence even a large volume mismatch—in our final null, 100 mL of tank water against a 5 L reference—produces *zero* differential error provided both vessels reach *full* equilibrium. The familiar "equal volumes for symmetry" rule applies only under *partial* equilibration, where unequal volumes sit at different points on the first-order approach curve $C(t) - C^* = (C_0 - C^*)\, e^{-t/\tau}$. The true requirements are just (a) both fully equilibrated and (b) matched p_{CO_2} and T. The larger volume is rate-limiting, with a time constant

$$\tau \;\propto\; \frac{V}{K_H\, Q_{\mathrm{gas}}}\,\beta \;\sim\; \frac{1}{\mathrm{VVM}}\,\beta, \tag{2}$$

where $\mathrm{VVM} = Q_{\mathrm{gas}}/V$ is gas volume per liquid volume per minute and β is a seawater chemical factor (only trace $[CO_2]_{aq}$ is strippable; the Revelle buffering raises τ by $\sim$ 8–15$\times$). Equation (2) is the basis for the "increase aeration to shorten the measurement" lever discussed later, and for the choice to miniaturize the rate-limiting reference.

2.5 The differential dKH formula

The firmware (Arduino Nano / ATmega328P; pH via DFRobot Gravity probe + ADS1115 16-bit ADC) digitizes one probe, reading reference then tank pH, and computes

$$\boxed{\mathrm{KH}_{\mathrm{tank}} \;=\; \mathrm{KH}_{\mathrm{ref}} \cdot 10^{-(\mathrm{pH}_{\mathrm{ref}} - \mathrm{pH}_{\mathrm{tank}})} \;=\; \mathrm{KH}_{\mathrm{ref}} \cdot 10^{\Delta \mathrm{pH}}} \tag{3}$$

with $\Delta\mathrm{pH} \equiv \mathrm{pH}_{\mathrm{tank}} - \mathrm{pH}_{\mathrm{ref}}$ and $\mathrm{KH}_{\mathrm{ref}} = 8.448\,\mathrm{dKH}$ an EEPROM-stored *anchor* constant (set by a `calref` command after evaporation top-off; *not* a per-run measurement—this is why the reference column reads a constant 8.448 in every log line). Equation (3) depends only on ΔpH; absolute pH cancels exactly. Differentiating,

$$\left.\frac{d\,\mathrm{KH}_{\mathrm{tank}}}{d\,\Delta \mathrm{pH}}\right|_{\Delta \mathrm{pH}=0} = \mathrm{KH}_{\mathrm{ref}} \ln 10 \approx 19.45\ \mathrm{dKH}\,\mathrm{pH}^{-1}, \tag{4}$$

i.e. 0.01 pH of uncancelled difference maps to $\approx$ 0.19 dKH. This high sensitivity is what makes both the cancellation and the plateau criterion matter. Temperature compensation is applied per reading by a Nernst relation $\mathrm{pH}_{\mathrm{corr}} = 7.0 + (\mathrm{pH}_{\mathrm{raw}} - 7.0)\,(273.15 + T_{\mathrm{cal}})/(273.15 + T_{\mathrm{meas}})$, nominally zero-error over 15–35 °C. Each pH reading is a trimmed mean of 64 samples (discard the top and bottom 16, average the middle 32) to reject aeration-bubble spikes.

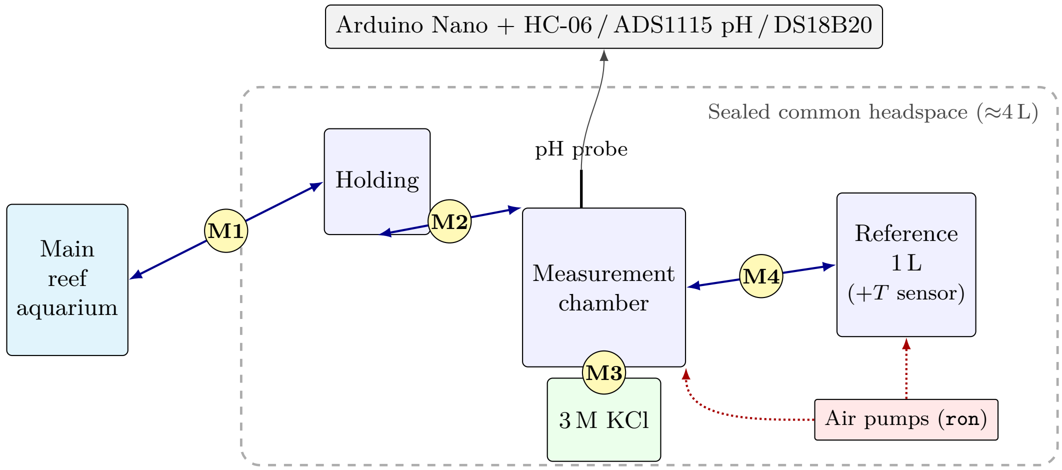

Figure 1: Schematic of the sealed closed-loop monitor. A single pH probe in the measurement chamber reads the tank aliquot and the reference in turn; four bidirectional peristaltic pumps (M1–M4) move liquid between the main aquarium, holding beaker, measurement chamber, reference reservoir, and KCl store; two air pumps (`ron`) aerate the reference and the chamber simultaneously. The reference, chambers and pump intakes share one sealed headspace (≈4 L), so sample and reference see identical p_{CO_2} and evaporation is suppressed; the controller sits outside. Between runs the probe rests in 3 M KCl (M3).

3 Apparatus

The build (Table 1, Fig. 1) is a closed fluidic loop around four bidirectional peristaltic pumps, two USB air pumps, a single pH probe, and a 1-Wire temperature sensor, all driven by an Arduino Nano over a 9600-baud Bluetooth (HC-06) serial link to a host PC.

Fluidic topology (current). All pumps are bidirectional (`f` = forward, `b` = reverse):

- **M1**: main aquarium ↔ *holding* beaker;
- **M2**: holding ↔ *measurement* chamber;
- **M3**: 3 M KCl beaker ↔ measurement chamber (probe soak / storage);
- **M4**: 5 L reference ↔ measurement chamber.

Routing the tank sample through an intermediate holding beaker lets it be temporarily *parked* while the reference is measured, minimizing the between-phase delay (Sec. 4). Run times are encoded as the integer in `mXf:NN` / `mXb:NN` (seconds); reverse runs ∼8–10 s longer than forward to fully clear the hose (current values `m1f:70`/`m1b:82`, `m2f:60`/`m2b:68`, `m3f:60`/`m3b:68`, `m4f:60`/`m4b:70`; the 5 L hose is longest, hence `m4b:70`).

Control commands. Air and motor enables are independent lines: `ron` turns *both* air pumps on (used only while liquid is stationary—bubble cleaning and measurement); `ton` enables the PWM motor controller (*required* before any pump will turn); `airoff` drops both. A load-bearing invariant follows: because `airoff` also disables the PWM controller, every liquid move must be preceded by `airoff` → `ton`; the last air command before a motor must be `ton`. Measurement primitives are `tank` / `ref` (sample and store one pH reading) and `calkh` (apply Eq. (3)).

Sealed common headspace (required). The 5 L reference, the measurement chamber, and the pump intake share *one* sealed headspace. This is not an optional refinement: it (i) guarantees tank and ref see identical p_{CO_2} at the instant of measurement (the differential's premise), (ii) blocks

Table 1: Bill of materials (current build).

Subsystem	Part	Qty
Microcontroller	Arduino Nano V3.0 (ATmega328P)	1
pH probe	DFRobot Gravity pH V2 (SEN0161-V2), single glass electrode	1
ADC	DFRobot Gravity I^2C ADS1115 (16-bit)	1
Temperature	DS18B20 waterproof (1-Wire), in the 5 L reference	1
Comms	HC-06 Bluetooth, 9600 baud	1
Pumps	NKP-DC-B06B 12 V peristaltic, bidirectional (M1–M4)	4
Motor drivers	L298 2 A modules (2 for pumps, 1 for air)	3
Air pumps	USB 5 V, switched together (`ron`)	2
Reference reservoir	5 L sealed container ("Wiz tank")	1
Chambers	200 mL beakers: holding, measurement, 3 M KCl	3
Probe storage	3 M KCl (≈22 % w/v)	—

room-CO_2 excursions at the source, and (iii) suppresses evaporation. In the final configuration the headspace is ≈4 L, the reference ≈1 L, and the tank aliquot 100 mL; an air stone placed *in* the reference actively pins the shared p_{CO_2} within minutes (the "anchor", Sec. 4). Separate or isolated headspaces are explicitly forbidden—two headspaces drift independently and destroy the cancellation.

4 Methods

4.1 Two-phase plateau-following cycle (V4)

Plateau detection is a host-side decision loop, so the PC script drives the firmware step by step. The cycle measures **tank first, then ref**, so the 5 L reference is co-aerated throughout the tank phase and converges quickly when its turn comes:

1. *Prep:* `m3b` (drain KCl from chamber) → `m1f` (aquarium → holding) → `m2f` (holding → chamber).
2. *Phase A:* `ron`; measure `tank` every 30 s until plateau (chamber tank + 5 L aerated together).
3. *Transition:* `m2b` parks the tank aliquot back in holding; `m4f` brings reference into the chamber.
4. *Phase B:* `ron`; measure `ref` until plateau (fast—already near equilibrium).
5. `calkh`; then cleanup: `m4b` (recover ref to 5 L), `m1b` (return parked tank to aquarium), `m3f` (restore KCl soak), `airoff` (probe rests in KCl).

4.2 Two-gate plateau criterion

Readings are held as integer milli-pH (mpH) to avoid floating-point boundary jitter. A plateau is declared only when *both* gates hold over a sliding window (Table 2): a short *span* gate (last 4 readings, max − min ≤ 2 mpH) rejects noise, and a longer *net* gate (last 8 readings, $|w_{-1} - w_0| \leq 1$ mpH) rejects a slow monotonic exponential tail that the span gate alone would mistake for convergence. The longer net look-back (8 vs. 4 points; "B1") was the fix for early latches in which the tank stopped climbing prematurely. Per-phase wall-clock and read-count caps (`PHASE_MAX` = 7200 s, `MEAS_MAX` = 240) and a consecutive-failure cap (`FAIL_MAX` = 5) bound the loop so it can never hang.

Figure 2 shows a measured pair of approach curves. The tank, freshly drawn into the chamber, climbs a slow first-order tail toward its ceiling and is held by the net gate until it is genuinely flat

Table 2: Plateau-following parameters (`measure_until_flat`).

Constant	Value	Meaning
`FLAT_SPAN_N`	4	span window (jitter gate)
`FLAT_SPAN_MPH`	2	last 4: $\max - \min \leq 2\,\text{mpH}$
`FLAT_NET_N`	8	net look-back (drift gate; B1)
`FLAT_NET_MPH`	1	last 8: $\lvert w_{-1} - w_0 \rvert \leq 1\,\text{mpH}$
`MEAS_INTERVAL`	30 s	between readings (aeration continuous)
`PHASE_MAX_SECS`	7200 s	per-phase cap (2 h; 4 h total)
`MEAS_MAX`	240	per-phase read cap
`FAIL_MAX`	5	consecutive parse failures tolerated

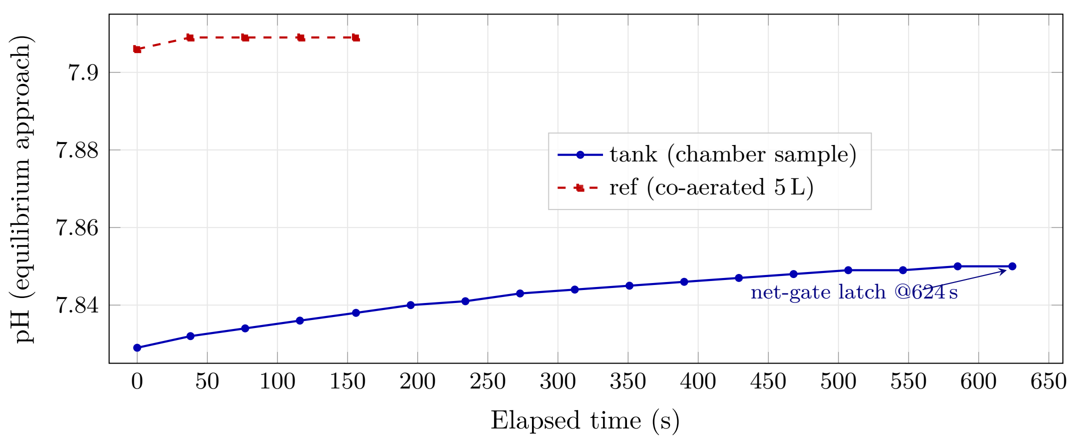

Figure 2: Measured plateau-following curves (sealed, net-gate run). The tank sample (solid) climbs a slow exponential tail and is latched only when the two-gate criterion (last-4 span $\leq 2\,\text{mpH}$ *and* last-8 net $\leq 1\,\text{mpH}$) is met; the co-aerated reference (dashed) is already near equilibrium and latches quickly. The two series sit at different pH because the underlying samples differ (this run predates the null campaign); only their respective *plateau* values enter Eq. (3).

(latching at $\sim$624 s when the 8-point net drift falls to 1 mpH). The reference, having co-aerated in the 5 L reservoir throughout the tank phase, is already at equilibrium and latches in $\sim$2.6 min—a direct illustration of why measuring tank-first is faster, and of why the span gate alone (satisfied near $\sim$546 s) would have latched the tank early.

4.3 Logging and fault handling

Each completed run appends one line `HH ref_pH tank_pH 8.448 tank_kh temp` to `dkh.dat`. Two distinct markers encode abnormal outcomes: an *all-zeros* line on hard failure (timeout, serial loss, or unverified KCl soak), which additionally *latches*—a subsequent run refuses to measure and re-writes the marker until a human clears it, protecting the probe from unattended pump cycling; and a *negative* `tank_kh` (magnitude preserved, sign flipped) when a phase hits its cap without reaching plateau, which the data server reads as "did-not-plateau" but which does *not* trip the latch. A `finally` block runs an emergency cleanup (air off, drain chamber, restore KCl soak) on any non-completion.

Table 3: Method/firmware revisions. “ron held” = continuous aeration between runs.

Rev	Period	Change vs. prior
V0	6/14–6/15	First symmetric build; reference drawn from 5 L into a cup; fixed 1200 s aeration.
V1	6/15–6/16	Added continuous aeration between runs (`ron` held).
V2	6/16–6/17	Reference taken as a small directly-aerated aliquot (holding → chamber); aeration off during the reading.
V3	6/17	Aeration *during* measurement (pin tank/ref to same p_{CO_2}); path symmetrized; KCl-rinse bubble clean (later dropped).
V4	6/17→	**Plateau-following** to true equilibrium (no fixed time); two-phase tank-then-ref; anti-hang caps; emergency cleanup. Subsequently: *sealed common headspace*, *5 L active anchor*, and the two-gate (B1) criterion.

4.4 Null-test protocol

The diagnostic backbone is the *null test*: the reference and tank vessels are filled with the *same* water so the true alkalinity difference is zero and the correct output is exactly ΔpH = 0, dKH = 8.448. Two variants are used. In a *partial null*, only the small tank aliquot is replaced with fresh seawater while the aged 5 L reference is retained, so a change in ΔpH isolates the tank water's contribution. In a *complete null*, reference water is also placed in the tank chamber, so the water difference is zero *by construction* and any residual ΔpH is purely instrumental.

4.5 Revision history and the one-change-at-a-time rule

The build evolved through five revisions (Table 3). Changes were made one causal axis at a time, so that each measurement campaign tests a single mechanism; where two constants were changed together (e.g. stabilization window and convergence band), they were justified as the *same* causal axis (response completeness), not two independent variables.

5 Results

Table 4 lists the full campaign. Throughout, $\Delta\text{pH} < 0$ means the tank read more acidic than the reference (lower apparent alkalinity), and by Eq. (3) the raw error is $\text{dKH} - 8.448 = 8.448\,(10^{\Delta\text{pH}} - 1)$.

Table 4: Complete measurement campaign (29 runs). “abs pH” is the common-mode level (a room-CO_2 proxy within a single day only). Config abbreviations: S = sealed headspace, A = 5 L active anchor, N = net-gate, B1 = net look-back 8. **Null staging:** a fresh seawater batch is aliquoted at the start; PN = partial null (the 100 mL tank chamber alone is refilled with that fresh aliquot while the aged 5 L reference is kept, isolating the tank-water contribution); CN = complete null (reference water is then transferred into the tank chamber as well, so both vessels hold *identical* water and the water difference is zero by construction).

#	Date/time	Cfg	pH_{ref}	pH_{tank}	ΔpH	dKH	T	Conditions
1	06-14 20:12	V0	7.959	8.021	+0.062	9.741	26.7	first manual symmetric run
2	06-14 21:00	V0	7.997	8.072	+0.075	10.054	26.2	—
3	06-15 05:00	V0	8.057	8.077	+0.020	8.843	26.4	dawn but +; window open (ventilated), low CO_2
4	06-15 13:00	V1	8.050	8.042	−0.008	8.293	27.0	first continuous aeration

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Table 4 (continued)

#	Date/time	Cfg	pH_{ref}	pH_{tank}	ΔpH	dKH	T	Conditions
5	06-15 21:00	V1	8.109	8.091	−0.018	8.109	27.3	clean low-CO_2 point
6	06-16 05:00	V1	7.810	7.751	−0.059	7.362	27.5	dawn closed-room CO_2 spike (both abs pH fell ∼0.3)
7	06-16 13:00	V1b	8.091	8.063	−0.028	7.922	27.7	tighter window/band
–	06-16 21:00	—	*failed (0.000; power cut, port semaphore)*					
8	06-16 21:21	V2	8.042	8.021	−0.021	8.039	28.6	first V2 (manual)
9	06-17 05:00	V2	7.997	7.939	−0.058	7.386	27.8	V2 dawn test *failed*; regressed to V1
10	06-17 13:00	V3	8.109	8.091	−0.018	8.095	28.2	first V3; ref under-equilibrated (5 reads)
11	06-17 21:00	V4	8.113	8.071	−0.042	7.669	27.8	first V4 + trim-mean firmware; clean plateaus
12	06-18 05:00	V4	7.738	7.695	−0.043	7.657	28.2	**decisive:** dawn −0.043 ≈ evening −0.042 despite 0.4 abs-pH swing ⇒ CO_2-level independence
13	06-18 13:00	V4	8.081	8.055	−0.026	7.950	28.3	same low CO_2 as #11 yet −0.026; dKH +0.29 jump ⇒ baseline *not* constant (evaporation, open 200 mL day 5)
14	06-19 05:00	V4+S	7.976	7.968	−0.008	8.283	28.1	**seal verif. ①**: dawn dip collapses −0.05 → −0.008; plateau 33→4 reads
15	06-19 13:00	V4+S	7.957	7.947	−0.010	8.255	28.6	**seal verif. ②**: dawn→noon jump +0.29 → −0.03 (∼10×); evaporation removed
16	06-19 21:00	V4+S	7.969	7.968	−0.001	8.421	26.8	near-perfect null (raw err −0.027, smallest); but +0.166 vs. #14/15 ⇒ offset not a perfect constant; AC transient (−1.8 °C)
17	06-20 05:00	V4+S	7.922	7.871	−0.051	7.521	27.7	dawn dip recurs; tank 11× monotone = early latch (not true plateau); AC cooling slowed kinetics
18	06-20 15:00	V4+SAN	7.975	7.929	−0.046	7.602	27.1	first 5 L anchor; tank *true* plateau (34×, 21.6 min) yet −0.046 ⇒ **genuine gap**, not latch artifact
19	06-20 18:11	V4+S,2×	7.979	7.924	−0.055	7.443	26.4	controlled exp; 2× aeration halves climb (1296→711 s) but ΔpH unchanged ⇒ *not* kinetics
20	06-20 18:57	V4+S,2×	7.975	7.931	−0.044	7.636	26.7	ref-first + hose swap; gap persists ⇒ order/path/non-equilibrium all rejected
21	06-20 19:50	V4+S	7.975	7.926	−0.049	7.539	26.9	reproduction
22	06-20 20:29	V4+SA	7.977	7.921	−0.056	7.426	26.9	tank-first+park; ref fastest ever (4×); tank early latch (7.921) ⇒ motivates B1
23	06-20 21:00	V4+SA,B1	7.979	7.926	−0.053	7.474	27.0	**B1 ok**: tank 25×/945 s true plateau; confirms #22 was ∼5 mpH short; gap (7th)
24	06-21 05:00	V4+SA,B1	7.974	7.928	−0.046	7.594	26.7	dawn matches evening cluster ⇒ time-of-day dependence ruled out (gap, 8th)
25	06-21 06:00	V4+S,PN	7.973	7.963	−0.010	−8.255	26.7	**partial null**: fresh tank only; tank pH +37 mpH, ΔpH −0.05 → −0.010; tank not plateaued (old cap) ⇒ lower bound
26	06-21 13:00	V4+S,PN	7.964	7.956	−0.008	8.298	27.2	**partial null, full eq.**: dKH 8.298, raw err −0.15; back in good cluster; 4 h cap validated
27	06-21 19:00	V4+S,CN	7.960	7.957	−0.003	8.387	27.4	**complete null**: reference water into tank chamber too ⇒ both identical; raw err −0.061 = pure instrument offset; 100 mL vs. 5 L cancels

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Table 4 (continued)

#	Date/time	Cfg	pH_{ref}	pH_{tank}	ΔpH	dKH	T	Conditions
28	06-21 21:00	V4+S,CN	7.959	7.958	−0.001	8.434	27.4	**complete null** (2nd): ΔpH − 0.001 (resolution limit), raw err −0.014 (smallest); reconfirms the ≤ 0.06 dKH offset
–	06-22 05:00	V4+S,CN			*failed (cap)*			55-min scheduler cap killed it mid-ref; tank had plateaued (7.955); fix: 5 h cap
29	06-22 09:00	V4+S,CN	7.959	7.948	−0.011	8.226	27.4	**aged complete null** (∼14 h, 100 mL exposed aliquot): null decayed to raw err −0.22; tank true plateau 59× (long aeration drives precipitation)

5.1 The dawn-CO_2 error and its diagnosis

The earliest clear pathology was a dawn collapse of ΔpH (run #6, −0.059; error −1.09 dKH). The decisive clue was common mode: *both* absolute pH values fell ~ 0.3 overnight (ref $8.109 \to 7.810$, tank $8.091 \to 7.751$), reversing the daytime upward trend—the signature of a closed-room CO_2 build-up acidifying the aeration water. A natural experiment confirmed it: on the one ventilated night (run #3, window open) the absolute pH stayed high and ΔpH was a normal +0.020, whereas closed nights gave $-0.058/-0.059$. Crucially, room CO_2 is exactly the common-mode variable the differential is *supposed* to cancel; its leakage into ΔpH signalled a broken ref/tank symmetry, not an environmental problem to be ventilated away.

5.2 Plateau-following and the inter-phase drift conflict

Measuring to a true plateau (V4) removed the CO_2-*level* dependence: runs #11/#12 gave ΔpH = −0.042 and −0.043 across a 0.4-unit swing in absolute pH. But the first V4 run had exposed a competing failure: a 31-minute tank measurement whose pH rose to a peak then *fell* (room CO_2 drifting during the measurement), latching at a value set by whenever the single probe happened to sample each vessel. This *inter-phase drift*, of order

$$\epsilon_{\mathrm{drift}} \;\approx\; \Delta t_{\mathrm{tank-ref}} \times \frac{dC}{dt}, \tag{5}$$

sets up a fundamental conflict with a single probe: full equilibrium needs long measurements, but long measurements widen the tank–ref sampling gap Δt and thus the drift. The same form with dT/dt explains run #16, where an air-conditioner cooling overshoot (−1.8 ℃ during the run) coincided with a +0.166 dKH excursion.

5.3 Sealing collapses both confounds

A single *sealed common headspace* resolves the conflict at the source. By compelling the reference and tank to share a single common air volume, it makes the common p_{CO_2} intrinsic rather than coincidental, blocks room-CO_2 excursions, and additionally suppresses the evaporation that had been concentrating the open tank aliquot. The effect was immediate (runs #14/#15): the dawn dip collapsed from −0.05 to −0.008; the dawn→noon dKH jump shrank ∼10× (from +0.29 open to −0.03 sealed); and the plateau time was reduced from ∼33 readings (∼50 min) to ∼4 (∼5 min)—which in turn collapses Δt in Eq. (5), structurally removing the inter-phase drift. The uncorrected error fell from −0.78 to ~ -0.18 dKH.

Table 5: Error budget after full diagnosis.

Component	Magnitude	Character / remedy
Aged-tank alkalinity loss	-0.86 dKH	sample handling; use fresh aliquot
Instrument offset	-0.06 dKH	ΔpH $= -3$ mpH; below correction value
Precision (1σ)	± 0.08 dKH	electrode noise + residual drift
Room-CO_2 dependence	≈ 0	removed by sealing
Evaporation confound	≈ 0	removed by sealing

5.4 The active anchor and the persistence of a genuine gap

Sealing alone left a slow headspace drift, because a passive shared air volume re-equilibrates only by surface diffusion (tens of minutes). Placing an air stone *in* the 5 L reference (the "anchor") actively pins the shared p_{CO_2} within minutes; a volume audit confirms the dominance hierarchy (sample $\sim$3 µmol $\ll$ headspace $\sim$65 µmol $\ll$ 5 L $\sim$500 µmol). Yet with the anchor and the net-gate in place, a ~ -0.046 to -0.053 gap *persisted* even when the tank reached a verified true plateau (run #18: 34 readings, 21.6 min). A controlled pair of experiments (runs #19/#20) then rejected, in turn, four device-side explanations: doubling the aeration halved the climb time but left ΔpH unchanged (*not* kinetics); reversing the measurement order and swapping hoses did not reverse the sign (*not* order, warm-up, or fluid path); and a tank started from a higher initial pH settled to the *same* equilibrium (*not* under-equilibration). The gap was a property of the *water*.

5.5 Null experiments localize the error

Because the device hypotheses were exhausted, we tested the water directly. A *partial null* (run #25, fresh seawater in the tank chamber only) collapsed ΔpH from -0.05 to -0.010 (tank pH $+37$ mpH); at full equilibrium (run #26) it gave dKH = 8.298, raw error -0.15—back in the good cluster. Swapping fresh water into the tank had recovered almost the entire error, implicating the aged tank aliquot. The *complete null* (run #27), with reference water in *both* vessels so the water difference is zero by construction, returned ΔpH $= -0.003$, dKH = 8.387, raw error $\mathbf{-0.061}$, with both sides at verified true plateaus despite the 100 mL:5 L volume mismatch—a direct confirmation of volume independence (Sec. 2).

5.6 Error decomposition and accuracy

The two nulls decompose the headline error (Table 5, Fig. 3):

$$\underbrace{-0.92\,\mathrm{dKH}}_{\text{raw}} = \underbrace{-0.86\,\mathrm{dKH}}_{\substack{\text{aged-tank alkalinity loss}\\ (\mathrm{CaCO_3}\text{ precip./biofilm})}} + \underbrace{-0.06\,\mathrm{dKH}}_{\substack{\text{instrument offset}\\ (\Delta\mathrm{pH}=-3\,\mathrm{mpH})}}\,. \tag{6}$$

Across the seven-point cluster (runs #18–24) the *precision* was ΔpH $= -0.0499 \pm 0.0048$ ($\approx$5 mpH), dKH $= 7.531 \pm 0.084$—a scatter about one-third of the ± 0.25 dKH goal. With the bulk error identified as a sample artifact and the instrument offset at -0.06 dKH (one-quarter of the goal), the monitor meets ± 0.25 dKH *uncorrected*; zero-point calibration was judged unnecessary, and with normal (non-degraded) reef water the uncorrected error should fall within ± 0.1 dKH.

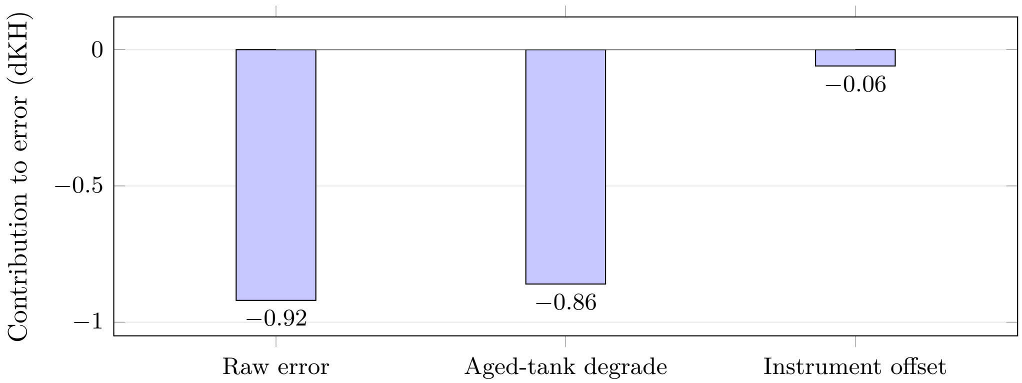

Figure 3: Decomposition of the −0.92 dKH raw error. The complete null isolates the instrument offset at −0.06 dKH (one-quarter of the ±0.25 dKH goal); the partial null attributes the remaining −0.86 dKH to alkalinity loss in the aged tank aliquot ($CaCO_3$ precipitation / biofilm)—a sample artifact, not an instrument fault.

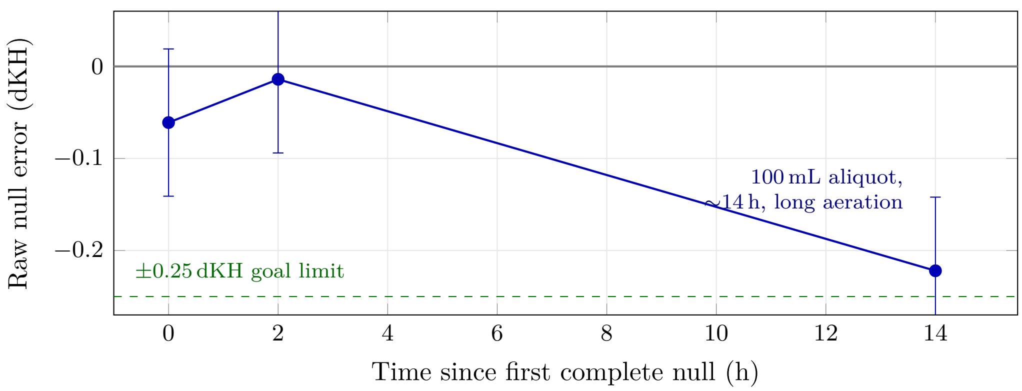

Figure 4: Time-resolved breakdown of the complete null. With reference water in *both* vessels (water difference zero by construction), the raw error stays within the ±0.08 dKH precision (error bars) and the ±0.25 dKH goal for the first hours, then falls to −0.22 dKH after the small (100 mL), exposed tank aliquot has aged ∼14 h—a long plateau-following run (59 readings) further driving $CaCO_3$ precipitation during aeration. The electronics are unchanged; only the sample alkalinity drifts, directly confirming that the dominant error term is sample chemistry (cf. Fig. 3).

5.7 Time-resolved breakdown of the null

The decomposition makes a falsifiable prediction: if the bulk error is sample chemistry rather than electronics, an otherwise-perfect null should *decay* as the sample ages. We tested this by re-running the complete null on the same 100 mL aliquot the next day, without touching the instrument (Fig. 4, run #29). The raw error—−0.061 and −0.014 dKH at 0 and 2 h—fell to −0.222 dKH at ∼14 h (ΔpH = −0.011; tank pH 7.957 → 7.948 while the larger reference held at 7.959). The drop is well beyond the ±0.08 dKH precision and was reached at a verified true plateau (tank 59 readings), so it is genuine alkalinity loss, not a latching artifact. The small, exposed aliquot and the unusually long aeration—which raises carbonate saturation and thus drives $CaCO_3$ precipitation *during* the measurement—are exactly the worst case for the −0.86 dKH mechanism, and the null broke down accordingly while the instrument offset floor stayed at its fresh value. This is the time-domain confirmation of the error decomposition, and a practical caveat: an anomalously long plateau on a small sample is itself a degradation warning.

6 Discussion

Implications of the error decomposition. The natural response to a stable $\sim -0.8\,\mathrm{dKH}$ offset is to zero-point calibrate. Resisting that—on the methodological principle that a calibration constant should be applied only after independent replication, and never to a small non-independent sample—was decisive, because the offset turned out *not* to be a constant device property (it varied $\pm 0.085\,\mathrm{dKH}$ across the sealed cluster and ultimately proved to be mostly degraded sample water). Calibrating it away would have embedded a sample artifact in the instrument's calibration and masked the genuine $-0.06\,\mathrm{dKH}$ floor.

External validation. Our independently derived error structure mirrors published behaviour of the commercial AquaWiz, lending it external support. Community reports identify room-CO_2 variation as the dominant error and a sealed container as the field fix; the manufacturer's own guidance—that a sharp CO_2 change "only affects readings for about an hour" and that one should "connect the air intake to the skimmer's outdoor air line"—matches both our $\sim$1-hour inter-phase drift episode and our gas-reference-fixing remedy (we realize it by sealing rather than by an outdoor line). The second-most-reported commercial error, bubble-quality degradation from salt creep ($> 2\,\mathrm{dKH}$ error, clean every six months), matches our aeration-sufficiency analysis. Reported field accuracy ($\leq 0.2\,\mathrm{dKH}$ vs. a Hanna checker; $\leq 0.75\,\mathrm{dKH}$ uncalibrated over seven weeks; "under $0.5\,\mathrm{dKH}$ = excellent, use for trend") brackets our $\pm 0.25\,\mathrm{dKH}$ result and our trend-oriented intent. Notably, commercial units are managed by periodic calibration and cleaning and do not pursue a true equilibrium plateau; our plateau-following step extends this further.

Scientific grounding. The kinetic separation underlying the method is well established: the aqueous CO_2 hydration/dehydration relaxation completes in $\sim$16 s [2], so chemistry is never rate-limiting; the slow step is physical gas exchange, $\sim$15 min for direct bubbling [5], consistent with our observed plateau times. The differential "A_T from $p_{\mathrm{CO_2}}$-equilibration plus pH" principle has a recent analytical-chemistry basis demonstrated down to 0.5 mL samples at ± 1–$2\,\mu\mathrm{mol\,kg^{-1}}$ precision [6], supporting small-volume miniaturization as the correct direction.

Limitations. (i) With a single probe the inter-phase drift of Eq. (5) cannot be driven to zero; it is only suppressed (by sealing and by minimizing Δt). (ii) The dominant historical error was sample degradation, so the operating discipline—use a fresh, adequately sized tank aliquot and recirculate rather than store—is part of the instrument, not an afterthought. (iii) Long plateaus under slow drift can make a measurement run to hours; the wall-clock caps bound this but a hard external scheduler limit must be set above the code's 4-hour ceiling. (iv) Absolute pH is informative only as a within-day CO_2 proxy; only ΔpH is quantitative.

Recommended operation. Seal the common headspace; anchor the reference with active aeration; measure tank-then-ref with the two-gate plateau criterion; use fresh sample water; clean the air path periodically; and treat the output as a trend instrument calibrated, if at all, against an independent kit on a monthly cadence keyed to the ref$\leftrightarrow$tank divergence.

7 Conclusion

A reagent-free, single-probe differential-pH aeration monitor can measure reef carbonate alkalinity to better than $\pm 0.25\,\mathrm{dKH}$ *without* zero-point calibration, once three design rules are observed: a single sealed common headspace, plateau-following to true gas equilibrium, and an actively aerated reference anchor. The value of the eight-day campaign was less in any one number than in *decomposing* a -0.92 dKH raw error: null experiments showed that -0.86 dKH of it was alkalinity loss in a degraded sample aliquot and only -0.06 dKH was the instrument. The differential design delivered its promised common-mode rejection of room CO_2 and probe drift; what remained was chemistry in a beaker, not error in the electronics. We expect the design rules and the null-decomposition methodology to transfer to other low-cost alkalinity instruments.

Reproducibility

Firmware, the host measurement script (`measure_kh_once.py`), the raw log (`dkh.dat`), and the full measurement ledger are available in the project repository [1].

Acknowledgments

This project was completed thanks to the passion of Mango-miyeok of REEF FAMILY and all the participants of the DIY meetup.

References

[1] nowforyou (REEF FAMILY), "ReefKeeper: open differential-pH aeration alkalinity monitor (firmware, host scripts, measurement ledger)," project repository, `https://github.com/taeseokyi/reefkeeper`.

[2] K. S. Johnson, "Carbon dioxide hydration and dehydration kinetics in seawater," *Limnol. Oceanogr.* **27**(5), 849–855 (1982). doi:10.4319/lo.1982.27.5.0849.

[3] A. L. Soli and R. H. Byrne, "CO_2 system hydration and dehydration kinetics and the equilibrium CO_2/H_2CO_3 ratio in aqueous NaCl solution," *Mar. Chem.* (2002).

[4] R. E. Zeebe and D. Wolf-Gladrow, *CO_2 in Seawater: Equilibrium, Kinetics, Isotopes*, Elsevier Oceanography Series 65 (2001).

[5] T. Takahashi, "Methods for measurement of $p$$CO_2$ and total CO_2 in seawater" (carbonate-system methodology; direct-bubbling equilibration ∼15 min).

[6] K. Fleger, X. Liu, W. Berelson, J. Adkins *et al.*, "Total alkalinity measurements in small samples based on CO_2 equilibration and spectrophotometric pH," *Anal. Chim. Acta* (2025). doi:10.1016/j.aca.2025.344432.

[7] (Gas-stripping first-order equilibration; $\ln(C_0/C_t) \propto \phi/V$.) Roberts, *Geophys. Res. Lett.* (2005).

[8] C. S. Egleston, C. L. Sabine, F. M. M. Morel, "Revelle (buffer) factor of the seawater carbonate system," *Global Biogeochem. Cycles* (2010).

[9] Reef-hobbyist community field reports on the commercial AquaWiz alkalinity monitor (room-CO_2 error, sealed-container fix, bubble degradation, manufacturer guidance): Reef2Reef threads #1159272, #1073924, #1115439; UltimateReef #939234; Bay Area Reefers #40433.